

Review article

Advancing urban forest and ecosystem service assessment through the integration of remote sensing and i-Tree Eco: A systematic review

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ABSTRACT

Urban forests support the health and well-being of billions of people living in cities globally. To better manage urban forests, it is crucial to assess their ecosystem services. This systematic review analyzes two established urban forest assessment approaches—i-Tree Eco and remote sensing—which have developed independently but hold significant potential for integration. The review, comprising the years 2008 to 2022, evaluates the current status of both methods in assessing urban forest structure and ecosystem services, highlighting opportunities for synergy. The literature shows that while both approaches primarily focus on regulatory services, remote sensing offers more versatile tools for assessing a broader range of ecosystem services beyond i-Tree's standardized scope. Remote sensing holds potential to enhance i-Tree Eco by providing structural and location-specific data at scale, albeit with varying accuracies. Studies have shown that LiDAR data reliably derives tree height and crown width, and that, in combination with multispectral and hyperspectral imagery, it enhances species identification. Additionally, mobile, and terrestrial laser scanners accurately estimate diameter at breast height. However, gaps remain in using remote sensing to assess crown characteristics like crown missing and dieback, which, though not critical, are useful for enhancing ecosystem service estimates in i-Tree Eco. Despite the potential of remote sensing to automate urban tree inventories, limited research has shown its successful integration with i-Tree Eco. Future research should standardize remote sensing techniques for assessing tree crown health. Additionally, further work is needed on quantifying differences between remote sensing and groundbased measurements, with the aim of evaluating uncertainty levels and understanding how these uncertainties impact the reliability and usefulness of data for policymaking and planning.

1. Introduction

Urban forests are vital green assets that support sustainable and resilient cities by providing a range of ecosystem services (ES). These ES have been categorized into provisioning, regulating, supporting, and cultural services (MEA, 2005b). Provisioning services include direct resources such as food and fuel derived from urban trees (Cabral et al., 2017; Potgieter et al., 2019). Regulating services involve the regulation of ecosystem processes and the benefits derived from those processes. Urban trees enhance air quality by removing pollutants (Baró et al., 2014; Bottalico et al., 2017), although their impact is context-specific, effective at large scale but limited or negative at street level (Venter et al., 2024). They regulate the micro-climate (Liu et al., 2021; Moody et al., 2021; Venter et al., 2020), help mitigate climate change by sequestering and storing carbon (Friess et al., 2016; Martin et al., 2012),

and prevent flooding by intercepting and absorbing rainwater (Csete and Gulyás, 2021; C. Xu et al., 2020). Additionally, trees can act as noise buffers reducing the impact of urban noise pollution (Xu et al., 2022). Supporting services are fundamental ecological processes that support the production of other ES. For instance, urban forests offer critical habitat for various species (Drago et al., 2017; Sasaki et al., 2016). Cultural services from urban forests encompass non-material benefits, including aesthetic appeal (Hu et al., 2022), cultural and recreational opportunities (Baumeister et al., 2020; Elze and Banzhaf, 2022), and enhanced physical and mental health for city dwellers (Engemann et al., 2019; Sinha et al., 2021).

Conversely, urban trees are also associated with certain disservices which have a negative impact on humans (Roman et al., 2021). For instance, at street tree level, urban trees can reduce airflow and trap pollutants, potentially increasing their concentrations (Venter et al.,

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2024). Additionally, urban trees pose a risk of failure, especially with global change. Poor management (Masini et al., 2023), extreme weather events such as windstorms (Portoghesi et al., 2023) and typhoons (Sit et al., 2024) intensified by climate change can lead to widespread urban tree failure, resulting in infrastructure damage and public safety risks. Other urban forest disservices include the release of air pollutants like biogenic volatile compounds (BVOCs) (Baró et al., 2014), induced allergenic responses (Ng et al., 2021), and problems like leaf litter and obstruction of good view (Roy et al., 2012). Given their critical role in human well-being, it is necessary to periodically assess the state of urban forest ES and disservices in our towns and cities.

A comprehensive approach is required to assess the state of urban forest ecosystems, the supply of ES, and their impact on people because of the potential trade-offs introduced by human intervention, where an increase in one ES might result in the reduction of another (European Commission, 2014; Gómez-Baggethun & Barton, 2013). ES assessment supplies decision-makers with quantitative evidence to support their policy and management choices. For example, access to urban forests is heavily influenced by socio-economic factors (Pedlowski et al., 2002). Advanced valuation methods can quantify the economic value of urban trees (Barton, 2023), while techniques for detecting changes in forest composition (Hanssen et al., 2021) offer essential insights for city planning and green infrastructure management. Understanding the structure of urban forests and their ecosystem services helps cities prioritize the equitable distribution of trees and their benefits.

Several methods have been developed for evaluating the ES and disservices by urban trees. Generally, these methods require detailed data on the structure and condition of the urban forests (Nowak et al., 2008). These data serve as inputs for models to estimate the ES of urban trees based on mechanistic or empirical principles. Typical methods to collect the required information are either bottom-up, relying on field data collection, or top-down, relying on remote sensing (RS). i-Tree Eco, one of the most widely used modeling tools in the i-Tree suite (Lin et al., 2019), uses a bottom-up approach, quantifying structural attributes of urban forests and estimating various ES, including air pollution removal, carbon storage and sequestration, stormwater run-off reduction, building energy saving effects, and BVOC emission (Nowak et al. 2008). While other tools in the i-Tree suite, such as i-Tree Canopy and i-Tree Landscape, utilize top-down methods with aerial imagery and pre-loaded data layers, i-Tree Eco stands out for its comprehensive capabilities in assessing urban forest structure and ES. This makes i-Tree Eco a valuable resource for detailed urban forest analysis and ES quantification and one of the focal themes in this research. Additionally, specialized tools like i-Tree Streets for quantifying street tree benefits, i-Tree Hydro for analyzing water flow impacts, and i-Tree Cool Air for assessing temperature effects in urban areas are also a part of the i-Tree suite.

i-Tree Eco offers a standardized approach to urban forest assessment with freely accessible tools and comprehensive, peer-reviewed models. Its sampling procedure offers a cost-effective means of estimating ES across entire cities (Scholz et al., 2018). Additionally, it can estimate ES at individual tree level, facilitating equity assessments across urban areas of varying scales (Baró et al., 2019). Besides that, i-Tree Eco can assign monetary values to tree benefits, supporting urban green project funding (Lai et al., 2020) and can also simulate the ES provision resulting from future tree planting scenarios (Amini Parsa et al., 2019; Widney et al., 2016; Zanzi et al., 2021). However, i-Tree Eco's US-centric models may not adapt well to diverse ecological settings globally, leading to uncertainties (Tan et al., 2021; Yarnvudhi et al., 2021) and its reliance on sample plot data for city-wide estimates introduces potential inaccuracies (Wei and Cheng, 2022). Although the model supports spatial analysis, it overlooks spatial variations, such as differences in air pollution levels, by applying a uniform average value to all trees (Cimburova and Barton, 2020). Furthermore, i-Tree Eco underestimates actual urban forest benefits as it only evaluates a subset of ES (Lai et al., 2020).

Field-based tree inventories are a fundamental source of input data for a number of ground-based ES models including i-Tree Eco. Despite existing standardized methods for creating tree inventories, the possibility of subjectivity in these estimates still exists (van Doorn et al., 2020), leading to inaccuracies. Also, conducting field measurements has its potential drawbacks, such as requirement for skilled labour and time (Nowak et al., 2008), substantial associated costs, challenges in accessing certain plots, aggregated outcomes that lack citywide spatial representation, and potential sampling errors due to sample size (Alonzo et al., 2016), thus introducing potential biases in the results.

In addition to bottom-up tools like i-Tree Eco, top-down RS methods and geographic information systems (GIS) tools have become popular for mapping and modelling urban forest structure and dynamics. RS presents a broad spectrum of data for ES assessment, determined by factors including the platform being used (ground-based, airborne or satellite), the sensor employed (active or passive), and the level of detail it provides in terms of spatial, temporal, or spectral resolution. Urban forest research extensively uses aerial imagery for tree characterization, particularly in mapping tree crowns and species (Velasquez-Camacho et al., 2021), offering a viable alternative to satellite imagery due to its optical capabilities (Li et al., 2019). Further, use of RS imagery advances the development of spatial models, while time-series data enhances our understanding of the dynamics of urban forests (Lin et al., 2019). Meanwhile, the use of Light Detection and Ranging (LiDAR) technology provides three-dimensional (3D) data on forest canopy and individual tree crown structure, offering effective estimation of attributes such as canopy/crown height and cover either from point cloud data or derived products like canopy height models (Baines et al., 2020). By characterizing urban forest structure and condition, RS can be used for benefit assessment of urban forests at both landscape and individual tree level (Tavares et al., 2019). Thus, RS offers the potential for quick, cost-effective, and multiscale urban forest description, supporting or replacing field-based methods.

Currently urban forest research using RS and i-Tree Eco predominantly relies on either one or the other of these two approaches for ES assessment. Despite their prevalence, there is a lack of comprehensive and focused reviews that address these assessment methods and their potential integration, highlighting the need for, and novelty of such a review. Therefore, given the desirability of using RS generated inputs to enhance field-based methods, this review evaluates the current status of RS and i-Tree in urban forest ES assessment and subsequently explores the integration potential of RS data with the i-Tree Eco model, specifically focusing on the technological aspects of data collection through RS to parameterize i-Tree Eco. Key questions addressed include: (1) What is the current state of urban forest ES assessment using RS and i-Tree? (2) Are there studies demonstrating successful integration of RS data with i-Tree Eco? (3) Which tree attributes can be reliably derived from RS for incorporation into the i-Tree Eco model?

2. Methods

2.1. Literature search strategy

A set of search terms were identified by breaking the overall topic, 'Modelling Urban Forest Ecosystem Services using Remote Sensing and i-Tree Eco' into four distinct concepts and determining keywords and synonyms for each concept (Appendix A, Table S1): urban forest (concept 1), ecosystem services (concept 2), remote sensing (concept 3), i-Tree (concept 4). Additionally Boolean operators (AND & OR) and truncation (*) were used for constructing search queries. The Scopus database was accessed on 8th April 2023, to conduct a structured literature search. The search period was limited to the years 2008–2022. This 15-year time period was chosen to capture the significant growth in ES research following the 2005 Millenium Ecosystem Service Assessment (MEA, 2005a), and the 2007–2010 The Economics of Ecosystems and Biodiversity initiative (TEEB, 2008, 2010), which garnered

widespread interest in this field (Costanza et al., 2014).

To further narrow down the search, the search fields were restricted to title, abstract and keywords, English language, and only original research articles and peer-reviewed review articles were included in this review to ensure the reliability of the sources. Two separate search queries were developed (Appendix A, Table S2) and searches were conducted to retrieve literature on the two distinct methods employed in urban forest ES research, with the aim of assessing their current utilization and the potential for their integration. The search identified 527 articles using RS and 117 studies using i-Tree for ES.

2.2. Screening for inclusion

Titles and abstracts were screened to eliminate irrelevant studies. The following screening criteria were established to determine the inclusion or exclusion of articles:

- (i) The study area must be based in urban/peri-urban forests.
- (ii) The study must assess one or more specific types of ES (e.g., carbon storage and sequestration, air pollution removal, habitat provision, noise reduction, etc.).

- (iii) The study must make use of RS or i-Tree methods for ES assessment.
- (iv) The study must use RS methods for deriving tree attributes required for i-Tree Eco.

Each article should either meet criteria (i) (ii) and (iii) or (i) and (iv) ensuring that selected articles either study urban/peri-urban areas and assess ES using i-Tree or RS methods or focus on urban/peri-urban forests for deriving their structural attributes using RS. After screening titles and abstracts, 159 of the 527 articles on RS and 101 of the 117 articles on i-Tree for urban forest ES assessment were shortlisted for full-text screening, with 11 articles common to both sets of literature.

2.3. Full-text screening

After full-text screening, using the same criteria as in 2.2, 122 of 159 RS articles (11 reviews, 111 original) and 88 of 101 i-Tree articles (3 reviews, 85 original) were selected for analysis. Review papers were used to provide context and support findings. Ultimately, 196 original articles (111 RS, 85 i-Tree) were analyzed. Fig. 1 illustrates the article selection process.

Among the 111 RS articles, 78 focused on urban forest ES assessment,

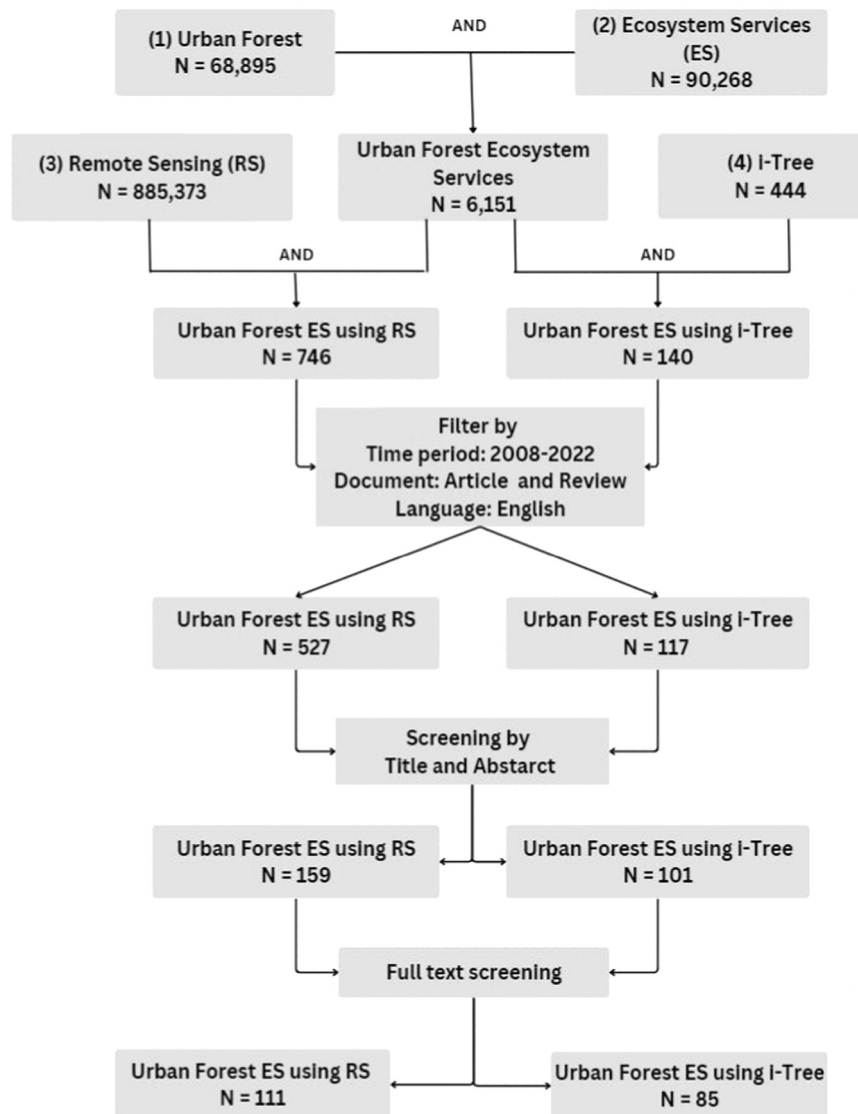


Fig. 1. Flowchart illustrating the systematic search and screening process for literature review article selection. Four search keywords, (1) urban forest, (2) ecosystem services, (3) remote sensing, and (4) i-Tree were searched in Scopus retrieving N number of articles for each search term and their combination.

21 on urban forest characterization, and nine on both. Additionally, three used RS and GIS methods to derive attributes for the i-Tree Eco model. In the i-Tree category, 82 of 85 articles assessed urban forest ES using various i-Tree tools, with three integrating RS methods: two parameterized the i-Tree Eco model, and one generated input for the i-Tree Cool Air model.

To answer research question (1) on the status of urban forest ES assessment using RS and i-Tree, 87 RS articles (78 + 9) and 82 i-Tree articles were analyzed. For research question (2) on the integration of RS data with i-Tree Eco, 13 studies were reviewed, and for research question (3) on deriving tree attributes from RS for the i-Tree Eco model, 30 studies (21 + 9) were examined.

3. Results and discussion

3.1. Current state of urban forest ecosystem service assessment using remote sensing and i-Tree

3.1.1. Remote sensing and i-Tree research trends

To examine the development of RS and i-Tree research compared to overall urban forest ES research, the total number of publications in these respective domains from 2008 to 2022 was assessed. As evident from Fig. 2, the overall urban forest ES research output has steadily increased in the past 15 years, starting from 53 articles in 2008, reaching 1039 articles in 2022. Similarly, urban forest ES research using RS and i-Tree exhibits an upward trend indicative of a field in progress although growth is less significant compared to overall urban forest ES research. Notably, the research literature using RS data began before and has grown more rapidly than the literature using i-Tree.

To visualize geographical trends, the shortlisted publications were classified by the country where the study was conducted (Appendix A, Table S3). Out of total 196 studies, 190 were based in a single country, four were conducted across multiple countries, one was a global study and one a simulation study. The global distribution of research was predominantly concentrated in the northern hemisphere. Fig. 3(a) depicts the geographic distribution of urban forest ES literature using RS

data, with the majority of research conducted in China, accounting for 30 %, the United States of America (USA) followed at 15 %, and Germany at 10 %. Canada and Italy comprised 4 % of the studies, while Australia, Norway, Chile, Colombia, Denmark, Japan, Mexico, South Korea and the United Kingdom (UK) each accounted for 2 %. On the other hand, as seen in Fig. 3(b), i-Tree has been widely adopted in urban forest ES research in the USA contributing to 29 % of studies. 13 % of the research on urban forest ES assessment using i-Tree originated from Italy, followed by 8 % from China, 5 % from the UK, and 4 % each from Germany, Japan, and Poland.

3.1.2. Urban forest ecosystem service assessment

The analysis of literature on urban forest ES assessment using RS and i-Tree reveals a strong focus on regulating ES (Table 1). Regulating services of urban forests were assessed 104 times in 87 articles using RS and 202 times in 82 articles using i-Tree. Notably, the total number of assessments exceeds the number of articles because a single study can evaluate multiple regulating ES. Using RS, the three most frequently evaluated services were carbon storage and sequestration (40 %), outdoor temperature regulation (34 %), and air quality regulation (22 %). For ES assessment using i-Tree, air quality regulation (87 %), carbon storage and sequestration (73 %), and water regulation (56 %) were the most frequently assessed services, among others. Regulatory ES have a direct and short-term impact on people (MEA, 2005b), are quantifiable, and directly influenced by the structure of the urban forest; therefore, they tend to attract more research attention. While both approaches emphasize the assessment of regulatory ES, RS technology offers a more versatile set of tools for evaluating a wide range of other ES categories (supporting, cultural and provisioning) provided by urban forests. It integrates diverse data from satellite, airborne, and ground-based sensors, operates across various spatial and temporal scales, and allows for customizable analysis methods, making it adaptable to specific research and monitoring needs (García-Pardo et al., 2022). This flexibility stands in contrast to the more focused approach of i-Tree, which is constrained by its structured methodology and a narrower focus on specific ES.

Cultural services, the second most frequently assessed ES were

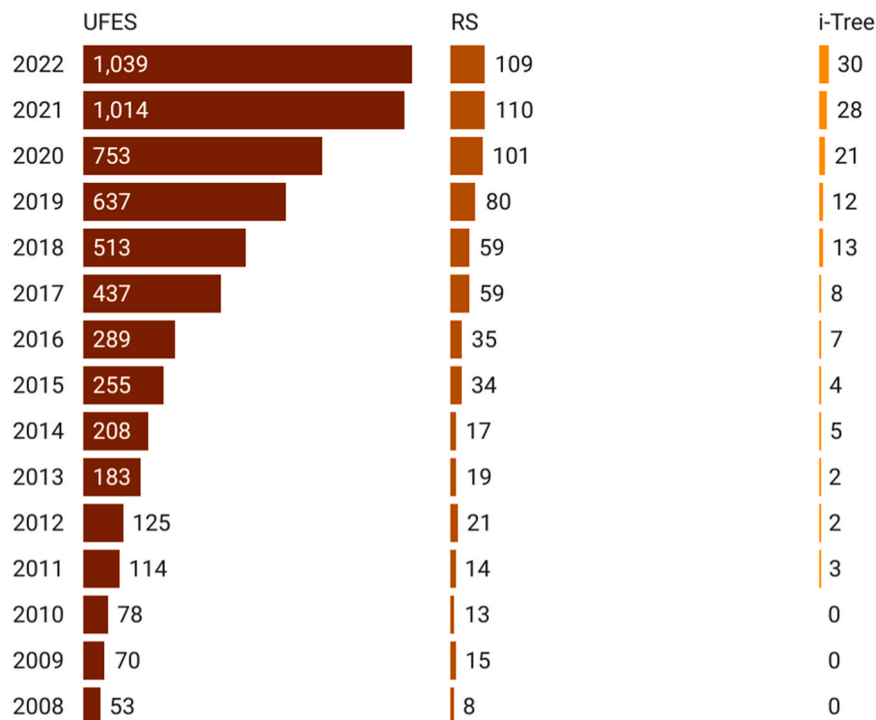


Fig. 2. Annual research output from 2008 to 2022 for overall urban forest ES (UFES) research and urban forest ES research using remote sensing (RS) and i-Tree tools.

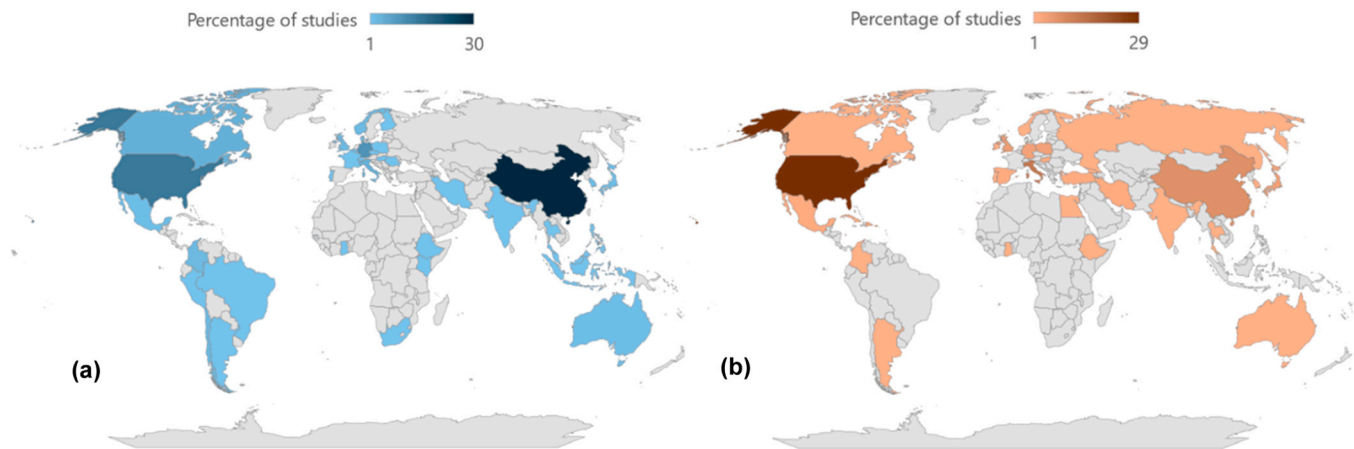


Fig. 3. Geographic distribution of urban forest ecosystem assessment research using (a) remote sensing and (b) i-Tree. The data are based on the countries where the studies were conducted. Higher percentage of studies is indicated by darker colour.

Table 1
Heatmap illustrating Provisioning (Pro), Regulating (Reg), Cultural (Cul), and Supporting (Sup) services, and Disservices (Dis) of urban forests, being evaluated using remote sensing (single and multi-source data) and i-Tree tools. Numeric values represent the number of times a specific ecosystem service or disservice was evaluated using remote sensing or i-Tree. Note that the total number of evaluations may exceed the number of articles because a single article can assess multiple ecosystem services. 6 studies that have used remote sensing data to parameterize i-Tree models have been excluded from the table to avoid double counting. References for studies can be found in [Appendix A, Table S4](#).

Urban Forest Ecosystem Services and Disservices		Remote Sensing																		i-Tree Tools						Total Number of Studies																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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		Satellite Lidar (SLS)	Airborne Lidar (ALS)	UAV Lidar (ULS)	Terrestrial /Mobile Lidar (TLS / MLS)	Aerial imagery* (Aer)	Satellite imagery# (Sat)	Satellite thermal imagery (SatTh)	ALS + TLS	ALS + Aer	ALS + Sat	ALS + Aer + Sat	TLS + Sat	ULS + Aer	Aer + Sat	Sat + Radar (SatRad)	Aer + SatTh	Sat + SatTh	Aer + Sat + SatTh	Sat + SatTh + SatRad	Aerial Thermal + Aer + ALS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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* Aerial imagery includes airborne/UAV RGB, multispectral, hyperspectral imagery, ranging from 0.1 to 5 m spatial resolution.
#Satellite imagery includes multispectral and hyperspectral imagery, ranging from 0.5 to 1000 m spatial resolution.

evaluated 22 times using RS and nine times using i-Tree. Using RS, recreation and health benefits of urban trees were assessed in 14 % of studies, followed by aesthetic and property value services in 8 %. Using i-Tree, aesthetic benefits of trees based on their impact on property values in cities were the most assessed cultural ES, accounting for 9 % of studies. Cultural ES have attracted less attention in ES research due to their subjective nature and the lack of commonly accepted method for assessing and valuing them, resulting in a predominant emphasis on material contributions in both research and decision-making (Chan et al., 2012). While mechanistic models such as i-Tree or RS technology have limitations in assessing cultural ES, recent studies suggest that they can still provide valuable insights into aspects such recreation and health (Elze and Banzhaf, 2022; Tan et al., 2021; Venter et al., 2020), aesthetic and property value (Dang and Li, 2021; Escobedo et al., 2015; Hu et al., 2022) and cultural heritage (Baumeister et al., 2020). However, social science methods remain essential for a comprehensive

evaluation (Tian et al., 2021).
Supporting services were addressed 10 times and provisioning services 4 times in the RS literature, but neither was assessed in any i-Tree studies. Supporting services, which have long-term and indirect impacts on people (MEA, 2005b), are typically omitted from ES assessments to avoid 'double counting' because their effects are reflected through their influence on provisioning, regulating, and cultural services (Haines-Young and Potschin, 2013).
In the limited studies on ecosystem disservices, BVOC emissions were examined in one RS study and seven i-Tree Eco studies, while allergenic responses were addressed in two RS studies but not addressed by i-Tree Eco.
The ES assessed in 169 articles (87 RS and 82 i-Tree) are summarized in Table 1 and the list of studies is available in Appendix A, Table S4. Table 1 details all ES evaluated, along with the specific RS data types and i-Tree tools used for each assessment. The following sections

compare urban forest ES research within the literature of each method, focusing on the most assessed ES—regulating services.

3.1.2.1. Regulating ES.

(i) Air quality regulation

RS and i-Tree are both used to assess the role of urban forests in regulating the quality of air by capturing atmospheric pollutants. i-Tree evaluates the air quality improvement by urban vegetation through the removal of carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), particulate matter smaller than 10 microns (PM₁₀), particulate matter smaller than 2.5 microns (PM_{2.5}), and sulphur dioxide (SO₂) (Nowak, 2021). For quantifying pollution removal, the most frequently used tool was i-Tree Eco (50 studies), followed by i-Tree Canopy (13), i-Tree Streets (7), and i-Tree Landscape (1).

These assessments, using i-Tree tools were conducted at different scales: 32 studies at the city level, seven at neighbourhood, and 14 at the street tree level. Additionally, 18 studies assessed trees in specific sites such as parks, university campuses, and city squares. One study was conducted in experimental setup at a nursery to estimate leaf area index (LAI) using i-Tree Eco and applied pollution deposition equations to assess PM₁₀ and O₃ removal (Baraldi et al., 2019).

In RS literature, satellite imagery was the most frequently used data source (11), followed by terrestrial or mobile laser scanning (2). Multi-source approaches included aerial imagery combined with ALS (3) or satellite imagery (1). Satellite imagery was also integrated with TLS (1) and with both ALS and aerial imagery (1). Remote sensing-derived LAI was often combined with deposition models and geospatial analysis to estimate air quality regulation (Bottalico et al., 2017; Manes et al., 2016; Marando et al., 2016). Pollution removal was also quantified using RS-derived vegetation volume combined with pre-established conversion factors (Fan et al., 2021; Gong & Chang, 2022; F.X. Li et al., 2022; Sun et al., 2022) and using RS-based land cover assessments with coefficient-based methods based on existing literature (de la Barrera et al., 2016; Elderbrock et al., 2020). More advanced approaches integrated low-cost air pollution sensors with dry deposition models, enabling species-specific calculations of air pollution removal rates (Kaspar et al., 2022).

(ii) Carbon storage and sequestration

Urban areas are major sources of CO₂ emissions and urban forests can offset some of these emissions. Carbon storage and sequestration assessments were mainly performed using i-Tree Eco (43), i-Tree Canopy (9), and i-Tree Streets (8). Carbon dynamics in urban areas were evaluated at citywide scale by 31 studies and neighbourhood scale by four studies. Nine studies specifically examined street trees for carbon storage and sequestration. Additionally, 16 studies focused on specific sites such as parks and university campuses.

Since stored carbon is directly related to and calculated from above-ground biomass (AGB) (Nowak and Crane, 2002), and serves as a critical indicator of carbon sequestration (Zeng et al., 2022), studies that account for AGB using RS techniques were considered under the ES of carbon storage and sequestration. Satellite imagery was the dominant single source RS data (12) followed by aerial imagery (3), ULS (2), ALS (2) and TLS (2) for carbon assessments. Multi-source approaches included ALS with aerial imagery (3), satellite imagery (2), TLS (1), and combined satellite and aerial imagery (1). Other data fusion approaches combined satellite data with thermal satellite imagery (3), aerial imagery (2), and satellite radar data (1) for evaluating carbon storage and sequestration. One study additionally used point

cloud data acquired through ULS and generated via Structure from Motion (SfM) photogrammetry (Lin et al., 2022).

(iii) Water regulation

Urban green spaces play a significant role in intercepting rainfall, facilitating water infiltration into the soil, consequently contributing to reduced run-off, improved water quality, and flood mitigation. Hydrological benefits of urban trees were primarily estimated using i-Tree Eco (26), though i-Tree Streets (8), i-Tree Canopy (6), i-Tree Hydro (5), and i-Tree Landscape (1) were also employed.

18 studies evaluated annual runoff reduction for urban forests citywide, while six studies estimated these benefits at the neighborhood level. 10 other studies specifically assessed the hydrological benefits of street trees, while 12 focused on specific sites such as parks, university campuses, and city squares.

For assessing the hydrological benefits of urban trees using RS, satellite imagery was the most common single data source (8), followed by aerial imagery (1) and TLS (1). Multi-source approaches included ALS combined with aerial imagery (2), a fusion of aerial and satellite imagery (1), and aerial imagery paired with satellite data (2). Using RS approach, several studies indirectly assessed water regulation services by using RS to map and classify urban green infrastructure, relating these to ES provision potential (Calderón-Contreras and Quiroz-Rosas, 2017; Gong and Chang, 2022; X. Li et al., 2021). Others used more direct methods, such as Baptista et al. (2018), who used TLS to correlate canopy metrics with water storage capacity and throughfall distribution. Modelling approaches using RS data as inputs for advanced models to simulate rainfall interception and run off reduction (Heris et al., 2021; Maragno et al., 2018; Mexia et al., 2018; C. Xu et al., 2020) and spatial analysis techniques (Elderbrock et al., 2020; B. Wang et al., 2019; J. Wang et al., 2019) were also employed for assessing water regulation potential. Additionally, multi-temporal analyses were conducted using time series satellite data to evaluate change in water regulation services over extended periods (Wangai et al., 2019; Y. Xiao et al., 2022; S. Xiao et al., 2022).

(iv) Indoor and outdoor temperature regulation

Cities commonly experience the urban heat island effect due to their high concentration of buildings, impervious surfaces, reduced vegetated areas and increased human activities (Kim, 1992). i-Tree has been used to assess temperature regulation by urban forests, with 24 % of the literature focusing on indoor temperature regulation and energy savings, though its application for evaluating outdoor heat mitigation is less common (6 %). Depending on their location and health, trees can reduce energy consumption in nearby buildings by providing evaporative cooling, shade, and windbreak (McPherson and Simpson, 1999; Nowak et al., 2017). The seasonal impact of urban trees on annual building energy consumption was evaluated using i-Tree Eco in 10 studies. Additionally, i-Tree Streets was used in seven studies, and i-Tree Canopy was used thrice.

Limited, research has estimated outdoor heat mitigation using i-Tree tools. Although i-Tree Eco cannot assess outdoor temperature regulation using canopy information (Scholz et al., 2018), two studies used i-Tree Eco to assess heat mitigation by employing tree transpiration as a proxy for its cooling impact on air temperature (Baró et al., 2019) and on the basis of modelling outcomes of gross carbon sequestration, net carbon sequestration, and carbon storage (Baró et al., 2014). In addition, Bodnaruk et al. (2017) coupled i-Tree Hydro with the physically based analytical spatial air temperature and humidity (Pasath) model to quantify air temperature and humidity. Lastly Nyelele

et al. (2019) used i-Tree Cool Air while Petri et al. (2019) used i-Tree Landscape to assess outdoor heat regulation.

In RS literature, the impact of urban forests on indoor temperature regulation and energy saving was addressed by only three studies (Chakraborty et al., 2022; Heris et al., 2021; Potgieter et al., 2019). In contrast, outdoor temperature regulation was often assessed using satellite thermal imagery (4). When satellite imagery (7) was used, it was typically combined with meteorological data (Rocha et al., 2022; Y. Xiao et al., 2022; Zhao et al., 2019), specific ES evaluation equations (Gong and Chang, 2022; Wan and Zhao, 2020; B. Wang et al., 2019), or vegetation indices to evaluate green spaces and their cooling effects (Calderón-Contreras and Quiroz-Rosas, 2017). Two other studies utilized aerial imagery to assess cooling benefits, either using carbon storage as a proxy for temperature regulation (Cabral et al., 2017) or employing microclimatic models such as ENVI-met and High-Resolution Vegetation Atmosphere Coupler (Lehmann et al., 2014). Multiple sources included the fusion of satellite imagery with satellite thermal imagery (11) and in six other studies, various data from passive sensors, such as aerial and satellite imagery, and active sensors, such as LiDAR and radar, were combined to assess the heat regulation effects of urban trees.

3.2. Top-down approaches in i-Tree

i-Tree incorporates a top-down approach primarily through its i-Tree Canopy and i-Tree Landscape tools. i-Tree Canopy uses aerial imagery and random sampling to assess land cover and tree benefits (Kim and Kang, 2022), while i-Tree Landscape provides macro-scale analysis using national datasets and map overlays to inform urban forest planning and management (Petri et al., 2019). However, RS and geospatial methods to improve i-Tree Eco by automating tree surveys and estimating their spatial environment, are scarcely integrated with i-Tree Eco (Cimburova and Barton, 2020). In the retrieved literature for both methods, 14 publications were identified that addressed RS approaches with i-Tree. Of these, 6 publications used RS imagery to assess current land cover (Kim et al., 2015; McPherson et al., 2011; Song et al., 2020) or land cover changes (Nowak et al., 2016; Puplampu and Bofo, 2021) followed by ES assessment using an i-Tree model or Soil Conservation Service Curve Number method, as in case of C. Xu et al. (2020). Additionally, 2 studies used i-Tree Eco inventory to train models for estimating urban forest structure (Baines et al., 2020) or validating RS-derived parameters (Alonzo et al., 2016). Finally, one used RS data to parameterize i-Tree Cool Air (Pace et al., 2022), while five studies used RS data to parameterize i-Tree Eco as discussed below.

Two studies combined RS data with i-Tree Eco to evaluate the pollution removal benefits of urban trees. For comprehensive ES assessments in i-Tree Eco, mandatory parameters include tree species and diameter at breast height (DBH), while accurate evaluation also requires tree height, crown size, crown condition, and specific location characteristics of the tree (Nowak, 2020). Zhao et al. (2018) used MLS data to derive individual tree attributes including tree height, crown width and DBH for street trees in Nantong, China and imported these into i-Tree Eco to model carbon sequestration and PM_{2.5} removal, with species data from field-observations. The structural attributes of trees for i-Tree Eco can be directly measured on the ground or estimated (e.g., LAI) within the model using these direct measurements (Nowak, 2020). Bypassing field measurements, Hirabayashi (2021) used Sentinel-2 LAI data in i-Tree Eco to assess urban forest air pollution removal in Tokyo, Japan.

Furthermore, three RS studies enhanced municipal tree inventories by using spatial data to address missing parameters for i-Tree Eco modeling. These studies derived both structural and location-specific tree attributes. In the first study, Cimburova and Barton (2020) enhanced the tree inventory of Oslo, Norway, by leveraging ALS and other spatial datasets, and increased the available dataset for i-Tree Eco

analysis from 19 % to 54 %. With ALS data, they effectively derived tree height and crown diameter (CD) using stem points from the municipal tree inventory. Where ground measured DBH was unavailable, it was predicted using allometric equations with ALS derived height and CD. This predicted DBH was then used to estimate height to crown base using species-specific allometric equations within i-Tree Eco. Notably, the authors used the available municipal species dataset and did not conduct additional species classification using RS data. Apart from structural attributes, i-Tree Eco additionally requires location specific details of the tree such as distance and direction of tree to buildings and crown light exposure (CLE), which is the facets of a tree exposed to sunlight and a vital parameter directly used to determine gross and net carbon sequestration and conditionally used for estimating carbon storage (Nowak, 2020). Cimburova and Barton (2020) estimated CLE by determining nearby structures within a buffer around the tree stem point and extracted pixels matching tree height using the canopy height model (CHM). Tangents were then drawn from the stem point to the segmented crown perimeter to calculate the CLE. Furthermore, they obtained the distance and direction of tree to the buildings nearby by measuring the distance and direction from the tree stem point of trees taller than 6 m to the three nearest residential buildings through a GIS procedure. CLE was also estimated by Scholz et al. (2018) using a GIS procedure with a high-resolution (0.5 m²) digital surface model (DOM1L) for Duisburg, Germany, to assess the cooling and pollution removal benefits of urban trees. Finally, Zięba-Kulawik et al. (2021) assessed ES of urban trees in Raciborz, Poland, by combining RS data with field measurements to estimate tree characteristics for i-Tree Eco. Using a CHM from ALS data and ground measurements, individual crowns were delineated, and tree height and crown area were determined. Subsequently, inventory data from 3692 trees was used to develop a statistical model predicting DBH from tree height and crown area. Tree species were randomly assigned based on their proportions from 41 reference plots.

3.3. Potential of remote sensing to parameterize i-Tree Eco

3.3.1. Urban forest characterization through remote sensing

A total of 30 publications focused on estimating urban tree characteristics for ES assessment. Urban forest research makes use of a wide variety of RS data for individual tree detection, species identification and measurement of structural tree attributes. While LiDAR data has been widely recognized for urban tree characterization, the majority of the studies use a fusion of active and passive RS technology due to their ability to provide spectral and spatial details through the point cloud. It is worth highlighting that field data has been acknowledged as the preferred source of supplementary information, particularly for validation of RS outputs and development of regression models. Table 2 summarizes the tree attributes, required by i-Tree Eco for ES assessment. Additionally, it identifies RS data used to derive these parameters, specifying the types of data used and the accuracy of the derived parameters. See Appendix A, Table S5 for the reference list.

(i) Individual tree detection (ITD)

Detailed individual tree mapping improves urban forest ES modelling accuracy by providing the individual tree data as required by i-Tree Eco, making tree identification the initial step when using RS methods. ITD remains challenging in dense urban areas with unclear crown boundaries, resulting in higher detection accuracy for trees along streets than in dense tree stands in parks (Münzinger et al., 2022). ITD in urban forests uses a combination of active and passive RS technologies. Wilkes et al. (2018) used ALS data to identify individual trees for AGB estimation across Greater London. Additionally, ALS was integrated with multispectral imagery in three studies: Hanssen et al. (2021) found that only 69 % of registered trees coincided with segmented canopy polygons, Münzinger et al. (2022) reported 54.3 % accuracy in tree detection, with higher precision along

Table 2

Summary of tree attributes required by i-Tree Eco, detailing attributes derived from remote sensing data in reviewed literature and their respective accuracies. See Appendix A, Table S5 for references.

Tree attributes required for i-Tree Eco	Remote Sensing Data				Field Data	Performance metrics	Reference
	Active		Passive				
	ALS	TLS/MLS	UAV LIDAR	Multispectral Data Platform, Spectral Range, Spatial Resolution			
Species				Ground, 204 bands (400-1000 nm), 512x512 pixels (31°x31° FOV)	✓	Overall Accuracy (OA) = 85%-96%	1
	22 pts/m ²			Airborne, AVIRIS 224 bands (365-2500 nm), 3.7 m	✓	Avg. of User's and Producer's accuracy = 37% - 96%	2
	25 pts/m ²			Airborne, 72 bands (363-1051 nm), 1 m	✓	OA (Hyper Spectral) = 51.1%; OA (ALS) = 61%, OA (Hyper Spectral +ALS) = 70%	3
				Satellite, WV2 (8+1 Panchromatic bands (400-1040 nm)), 0.5-2 m	✓	OA (ML) = 75.1%, OA (SVM) = 71%	4
	DSM (4pts/m ²) DTM (1pt/m ²) NPD* = 1 NPD* = 8			Satellite, RapidEye [®] (5 bands (440-850 nm)), 6.5 m Airborne, HyMap (125 (VIS, NIR, SWIR1, SWIR2) bands (450-2480 nm)), 3 m Airborne, orthoimagery (RGB), 0.09 m	✓ ✓	F1 Accuracy 63-100%, User's Accuracy = 66-99%, Producer's Accuracy = 60-100% Original Dataset OA: (RF) = 46.3%, (MCC) = 79.6% MNF Dataset OA: (RF) = 87%, (MCC) = 88.9%	5 6
DBH		Data1 Accuracy = 12.22 mm Data2 Accuracy = 5 mm			✓	R ² = 90.01	7
		Four datasets with point densities between 273.17pts/m ² -1234.08pts/m ²				Not mentioned	8
		Accuracy = 0.9 mm @ 50 m			✓	RMSE = 1.85-2.12 cm Mean Errors = -0.31–1.10 cm	9
Height	Not mentioned			Satellite, Sentinel-2 Level-2A (15 bands from 443-2190 nm), 10-60 m	✓	RMSE = 4.9-6.2 m	10
	NPS** = 1.25 m				✓	—	11
			2500, 1000, 500, 250, 100, 50, 25, 10 and 5 returns/ m ²		✓	RMSE = 1.62-2.27 m	12
			Scanning Accuracy = 5 cm	Airborne, RGB, 5472 × 3648 pixels	✓	R ² = 0.9341, RMSE = 0.59 m	13
	8pulses/m ² 4.6 pts/m ²			Airborne, orthoimagery (RGB), 0.10 m Airborne, orthoimagery (RBG+NIR), 0.2 m	✓ ✓	R ² = 0.89 R ² = 0.79, RMSE = 1.9 m	14 15
Crown size	8pulses/m ²			Airborne, orthoimagery (RGB), 0.10 m	✓	R ² = 0.90	14
			Accuracy = 5 cm	Airborne, RGB, 5472 × 3648 pixels	✓	R ² = 0.9006, RMSE = 0.28 m	13
Crown radius	NPS** = 1.25 m				✓	—	11
Crown width		Accuracy = 0.9 mm at 50 m			✓	RMSE = 0.24-0.37 m Mean Errors = -0.01–0.14 m	9
Crown width	4.6 pts/m ²			Airborne, orthoimagery (RBG+NIR), 0.2 m	✓	R ² = 0.83, RMSE = 1.4 m	15
Crown height		Four datasets with point densities between 273.17pts/m ² -1234.08pts/m ²				Not mentioned	8
Crown missing		This parameter was not estimated in the reviewed literature.					
Crown light exposure	DSM (0.5 m ²)					—	16
	DSM (0.5 m ²)				✓	—	17
Crown health (dieback)	This parameter was not estimated in the reviewed literature.						
Distance & direction to building	DSM (0.5 m ²)				✓	—	17

*Nominal Point Density (Pulses/m²)
**Nominal Pulse Spacing
^ Multitemporal images within a single growing season

AVIRIS: Airborne Visible Infrared Imaging Spectrometer
DSM: Digital Surface Model
DTM: Digital Terrain Model

FOV: Field of View
MCC: Multi Class Classifier
ML: Maximum Likelihood

MNF: Minimum Noise Fraction
NIR: Near Infrared
RF: Random Forest

RGB: Red Green Blue
RMSE: Root Mean Square Error
SVM: Support Vector Machine

SWIR: Short Wave Infrared
VIS: Visible
WV2: World View2

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streets (71.9 %) than in parks (30.5 %). Wallace et al. (2021) identified individual trees within 2 % of manual counts, with the segmentation algorithm matching 74–88 % of manually identified crowns. A higher accuracy (91 %) was achieved for tree crown delineation by Chi et al. (2020) by combining hyperspectral imagery with ALS. Using ULS data for a study in university campus, Gülçin and Konijnendijk Van Den Bosch (2021) reported Dalponte algorithm's highest accuracy with an F1 score of 0.83 for ITD.

For individual tree point cloud extraction using MLS data, Zhao et al. (2018) achieved extraction completeness rates of 97.9 %, 98.4 %, and 100 %, and correctness rates of 92 %, 98.4 %, and 93.2 %, for three street tree datasets respectively. Fan et al. (2021) on the other hand achieved F1 scores above 0.90 for ITD of street tree datasets from four different cities.

(ii) Species

Application of RS approaches in urban areas is challenging due to complex spatial and spectral patterns, buildings causing shadows and obstructions, rapid changes, and diverse vegetation types including individual trees (Tigges et al., 2013). Notably, urban tree species classification is often conducted on a broader scale, such as functional type (Z. Xu et al. 2020; Yan et al., 2018; Yang et al., 2019) or genera (Tigges et al., 2013). When performing species-level classification for individual trees in urban forests, the focus is typically on a limited number of the most common species. This approach highlights the challenges of mapping tree species in complex urban environments using RS techniques.

Abbas et al. (2021) used multiple bi-monthly ground-based hyperspectral imagery to analyze phenology patterns of 75 trees, identifying 19 species with 85–96 % accuracy. Molnár et al. (2020) used WorldView-2 multispectral imagery, demonstrating that maximum likelihood (ML) achieved superior overall accuracy (75.1 %) compared to support vector machine (SVM)

classifiers for classifying six urban tree species.

Other research coupled airborne hyperspectral imagery with ALS for species classification. Alonzo et al. (2016) combined hyperspectral data and LiDAR-derived structural metrics to map 29 urban forest species, achieving high classification accuracies, including 90 % for *Platanus racemosa*, 93 % for *Pinus pinea*, and 83 % for *Quercus agrifolia*. Liu et al. (2017) integrated LiDAR-derived crown structure with hyperspectral data for 15 tree species, improving species classification to 70 %, particularly benefiting evergreen trees and those with distinct crown structures. Similarly, Zhang et al. (2016) integrated LiDAR-derived crown segments with hyperspectral data, achieving enhanced classification accuracies of 87.0 % and 88.9 % using Random Forest and Multi-Class Classifier algorithms, respectively, with 20 optimal bands selected via Minimum Noise Fraction.

(iii) Diameter at breast height (DBH)

DBH is a crucial tree attribute, significantly contributing to the determination of above-ground biomass, which is important for evaluating the ES of urban trees. Directly estimating DBH using data derived through passive RS sources is not feasible. While active RS can penetrate the tree canopy to offer a 3D view of urban forests, obtaining this information requires dense point cloud data, which can be costly and complex to process. A number of studies relied on modelling DBH values using ALS- (Cimburova and Barton, 2020; Gülçin, 2021; Gülçin and Konijnendijk Van Den Bosch, 2021; Shrestha and Wynne, 2012; Tigges et al., 2017; Zięba-Kulawik et al., 2021) or ULS-derived (Zhang et al., 2022) height and/or crown metrics, or predicting the DBH values using NDVI-based models (Ren et al., 2017; Ren et al., 2015). This highlights the insufficiency of ALS and ULS data to measure DBH directly.

Ground-based laser scanners, which provide a higher point cloud density for assessing tree structure, have been employed to estimate the DBH trees with high accuracy. Guo et al. (2022)

observed TLS-derived DBH to strongly correlate with field-measured DBH ($R^2 = 90.01$). Zhao et al. (2018) analyzed street tree datasets from three streets using MLS to measure DBH, reporting RMSE between 1.85 cm and 2.12 cm, and Mean Errors from -0.31 cm to -1.10 cm. Fan et al. (2021) also used MLS point cloud data to measure DBH but did not report the accuracy of their assessment.

(iv) Height

Accurate measurement of tree height is crucial for assessing the benefits of urban forests. Baines et al. (2020) used ALS data, ground measurements, and Sentinel-2 NDVI to map canopy height in Greater London, reporting a mean height of 8.1–15.0 m with an RMSE of 4.9–6.2 m. However, similar studies suggest that NDVI is more effective in predicting urban vegetation attributes such as basal area and LAI rather than tree height and DBH (Ren et al., 2017; Ren et al., 2015).

In addition, Shrestha and Wynne (2012) achieved high accuracy using ALS data for tree height estimation, with an overall R^2 of 0.89, noting variations between conifers ($R^2 = 0.789$) and broadleaf trees ($R^2 = 0.889$). Similarly, Münzinger et al. (2022) found a strong correlation ($R^2 = 0.79$) between LiDAR and field-measured tree heights, but noted increasing underestimation for taller trees, with an overall RMSE of 1.9 m. Blackman and Yuan (2020) observed that tree height estimates varied by 0.91–2.43 m between CHM and point cloud segmentation methods.

More recently, Da Cunha Neto et al. (2021) evaluated ULS's ability to measure heights of *Araucaria angustifolia* trees, finding strong correlation with ground-based measurements at point density of 100 returns/m². Accuracy remained high even with reduced point densities down to 25 returns/m². Lin et al. (2022) leveraged point cloud data sourced from ULS and Structure from Motion (SfM) photogrammetry for tree heights, achieving high accuracy ($R^2 = 0.93$ and RMSE = 0.59 m).

(v) Crown size and condition

Tree crown size and condition significantly influences the functioning and ES-delivery. Crown radii/diameters can be derived directly from segmented crowns using point clouds or CHM, while estimating crown base height requires detailed point cloud data. ALS-derived crown radius and diameter have shown high correlation with field measurements, as observed by Shrestha and Wynne (2012) ($R^2 = 0.90$), and by Münzinger et al. (2022) ($R^2 = 0.83$) respectively. Blackman and Yuan (2020) on the other hand observed a significant variation between CHM and point cloud methods for crown width estimation, with CHM measuring 25.76 m and point cloud at 38.98 m for the same tree. Lin et al. (2022) assumed circular tree crowns and used ULS-derived CHM to measure crown radii. The radius ranged from 1.25 m to 2.72 m, averaging 1.98 m, with an R^2 of 0.9006 and an RMSE of 0.28 m. In addition, Zhao et al. (2018) assessed crown width of street trees using MLS data from three streets, reporting RMSE between 0.24 m and 0.37 m and mean errors from -0.01 m to -0.14 m.

Direct measurements of crown height were only reported by Fan et al. (2021), while Cimburova and Barton (2020), predicted crown base height using species-specific allometric equations from i-Tree Eco, based on DBH. However, none of the studies reported the accuracy of their estimation.

Additionally, no studies focused on assessing missing canopy or crown health characteristics such as dieback, as required by i-Tree Eco. These attributes typically rely on ground-based visual interpretation of tree crowns, making the assessment subjective. However, F.X. Li et al. (2021) used high resolution aerial imagery and TLS data to create a model for estimating 3D green volume trees, achieving high prediction accuracy of 88.07 % and Sun et al. (2022) estimated the effective living vegetation volume of

individual Ginkgo trees using TLS data. These types of estimations, when conducted at the individual tree level, show promising potential for effectively assessing crown size and condition in a standardized manner. Furthermore, Chi et al. (2020) emphasized on the critical role of ALS indices (pertaining to point density, tree dimensions, and morphology) and hyperspectral image indices (associated with chlorophyll absorption) in describing tree health.

3.3.2. Enhancing i-Tree Eco with remote sensing

i-Tree Eco performs ES assessment using detailed, time-intensive, and costly ground-based urban forest surveys. This limitation was also highlighted by Raum et al. (2019), who noted that the lengthy surveys in UK i-Tree projects were seen as a major barrier, potentially reducing the project impacts. Therefore, resource limitations and the complexity of quantifying ES through field surveys hinder the routine integration of urban tree ES into planning processes (Scholz et al., 2018). To this end, RS technology presents time and cost-efficient solutions for generating automated tree inventories, especially for trees to which surveyor access is limited (Zięba-Kulawik et al., 2021). The limited research integrating RS-derived input with i-Tree Eco modelling suggests that while RS may not completely substitute manual surveys, it can certainly enhance the quality and efficiency of collection of some input data used in i-Tree Eco (Cimburova and Barton, 2020; Scholz et al., 2018; Zięba-Kulawik et al., 2021).

RS techniques show considerable potential for deriving several key tree attributes required by the i-Tree Eco model, albeit with varying levels of accuracy. Individual tree detection, species classification, height, and crown estimation can be reliably derived using a combination of LiDAR and multispectral or hyperspectral imagery. These methods generally achieve good accuracy, with R^2 values often exceeding 0.90 for tree height and crown width measurements. While DBH cannot be directly measured through satellite or aerial imagery, it can be modelled using LiDAR-derived metrics or accurately measured using ground-based laser scanners including TLS and MLS. However, some attributes like crown base height and crown condition assessments still present challenges for RS-based derivation. Despite these limitations, the overall findings suggest that RS can provide valuable data for many of the tree attributes required by i-Tree Eco, potentially improving the efficiency and scale of urban forest and ES assessments.

4. Limitations of the review and future work

This review focuses on the technological aspects of urban forest ES assessment using RS and i-Tree. In this final section we highlight some limitations and areas for future research.

First, the search strategy used in this study may have excluded some relevant work, particularly grey literature outside the Scopus database and research published in non-English languages. This could limit the generalizability of the findings, especially when applied to influence policy in specific urban contexts. Furthermore, while academic advances in nature valuation methods are progressing, their practical integration into policymaking remains limited, partly due to methodological, resource, and institutional constraints (Pascual et al., 2023). Similarly, the accuracy of tree attributes derived through RS is highly context dependent. For example, broad-scale urban planning may allow lower precision in tree canopy estimates, whereas detailed ES valuations or individual tree management decisions require higher accuracy. Moreover, real-world policy implementations of RS data and i-Tree are commonly published in government documents or non-academic sources, often in languages other than English. As a result, the current academic literature provides limited documentation on how i-Tree and RS assessments are being used in policymaking.

To address these limitations, future research is needed to understand how urban forest ES are analysed and this information used in grey literature, government reports, and non-English studies, to provide a

more inclusive understanding of global practices. This would offer a clearer picture of how RS and i-Tree data are applied in policy contexts. Future research should also investigate which RS-derived tree attributes are accurate enough for policy purposes, such as urban planning, ES evaluation, and sustainability goal setting. Comparative studies between RS data and traditional field-based methods should examine uncertainty and how its propagation influences the reliability and usefulness of the information for policymaking and planning. Additionally, research should evaluate the cost-effectiveness and efficiency of these technologies in different decision-support scenarios. By addressing both technical and practical considerations, future work can better assess the applicability of RS data in i-Tree Eco for urban forest ES assessments.

5. Conclusion

Remote sensing offers an effective alternative to field-based measurements by enabling the quantification of forest structure across extensive urban areas. This data can then be used to enhance ground-based ES modelling tools, improving the scale and efficiency of urban forest assessments. Despite this, research specifically focussed on using RS to parameterize i-Tree Eco, a widely adopted urban forest analysis tool, is limited. The findings indicate that while both approaches emphasize assessing regulatory ES, RS technology provides a broader range of tools for evaluating various ES categories in urban forests, compared to the more focused and standardized approach of i-Tree, which accounts for a limited set of urban forest ES.

The literature indicates that not all i-Tree Eco input parameters have been estimated using RS techniques. However, LiDAR technology demonstrates high accuracy in determining critical tree attributes required by the i-Tree Eco, such as tree height and crown width. Although, combining LiDAR with multispectral and hyperspectral data, along with machine learning algorithms, improves species classification accuracy, none of the studies performed species classification for parameterizing i-Tree Eco. DBH, another vital parameter for i-Tree Eco, has mostly been modelled using the statistical dependencies between ALS-derived tree height and crown metrics in integrated studies, but has also been accurately estimated using ground-based LiDAR. Review results suggest that a combination of active and passive remote sensing technologies, coupled with sophisticated data analysis methods, offers a promising approach for deriving comprehensive and precise tree parameters, potentially improving the quality and efficiency of urban forest assessments using i-Tree Eco.

While this review provides valuable insights into the current state of RS and i-Tree applications in urban forest ES assessment, it also highlights several areas for future research that could further enhance the integration and practical application of these technologies in urban forest management and policymaking.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the first author used ChatGPT to improve readability and language. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRedit authorship contribution statement

Justin Morgenroth: Writing – review & editing, Supervision, Methodology, Conceptualization. **Daniel R Richards:** Writing – review & editing, Supervision, Conceptualization. **Gunjan Sharma:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Ning Ye:** Writing – review & editing, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ufug.2024.128659](https://doi.org/10.1016/j.ufug.2024.128659).

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